

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Data Engineering professional with **6+ years of experience** designing, building, and supporting enterprise-scale data solutions using Python, SQL, Apache Spark, PySpark, Microsoft Azure, Databricks, and modern cloud-native data platforms.
- Hands-on experience implementing **data migration, ingestion, transformation, curation, modeling, and consumption layers** using Azure Data Lake Storage, Azure Data Factory, Azure Functions, Event Hubs, Azure Stream Analytics, and Azure Databricks.
- Strong expertise in building scalable **batch and streaming data pipelines** using Python, PySpark, Spark SQL, Delta Lake, ETL/ELT patterns, event-driven architectures, and traditional data warehouse technologies.
- Experienced across the complete data development lifecycle including **solution design, cloud engineering, infrastructure, networking, security, ingestion, preparation, data modeling, performance tuning, deployment, BI consumption, alerting, and production support**.
- Worked closely with **source application teams, product owners, business stakeholders, and analytics teams** to translate complex data requirements into reliable analytical solutions that support operational insights and data-driven decision-making.
- Strong DevOps experience implementing **CI/CD pipelines, Infrastructure as Code, Git-based development, automated deployments, monitoring, alerting, and production operations** while owning data products throughout their lifecycle.
- Demonstrated technical leadership in Agile environments through **architecture decisions, reusable data-engineering patterns, performance optimization, mentoring, incident resolution, documentation, and cross-functional collaboration**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Data Processing:** Python, SQL, PySpark, Apache Spark, Spark SQL, Java, Scala Concepts, Pandas, DataFrames, Distributed Processing, Batch Processing, Stream Processing.
- **Microsoft Azure:** Azure Data Lake Storage (ADLS), Azure Data Factory (ADF), Azure Databricks, Azure Functions, Event Hubs, Azure Stream Analytics, Azure Storage, Azure Key Vault, Azure Monitor, Azure DevOps.
- **Data Engineering:** ETL/ELT, Data Migration, Data Ingestion, Data Preparation, Data Curation, Data Modeling, Data Integration, Data Warehousing, Delta Lake, Lakehouse Architecture, CDC, Structured and Streaming Data.
- **Databases & Analytics:** SQL Server, PostgreSQL, MySQL, Oracle, Delta Tables, Data Warehouses, Dimensional Modeling, Star Schema, Snowflake Schema, BI Consumption, Data Quality, Performance Tuning.
- **Cloud & Infrastructure:** Microsoft Azure, AWS, GCP Concepts, Cloud Infrastructure, Networking, IAM/RBAC, Security, Docker, Terraform, Infrastructure as Code, Environment Configuration, Cloud Administration.
- **DevOps & Operations:** Azure DevOps, CI/CD Pipelines, Git, Jenkins, TeamCity, GitLab CI/CD, Release Automation, Deployment, Monitoring, Logging, Alerting, Production Support, Incident Management, Agile/Scrum.
- **Data Platforms & Architecture:** Azure Databricks, Palantir Foundry, Data Lakehouse, Medallion Architecture, Event-Driven Architecture, Streaming Pipelines, Analytics Solutions, Data Governance, Consumption Layers, Technical Leadership.

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Architected and delivered enterprise **data engineering and analytics solutions** using Python, SQL, PySpark, Azure Databricks, Palantir Foundry, and cloud-native services, enabling insurance operations and product teams to derive actionable insights from large transactional datasets.
- Designed scalable **batch and streaming data pipelines** using Azure Data Factory, Azure Data Lake Storage, Databricks, Event Hubs, Azure Stream Analytics, Spark, and Delta Lake to ingest, transform, curate, and distribute structured and semi-structured enterprise data.
- Implemented complex **data migration, CDC, reconciliation, transformation, and data-modeling workflows** across source systems, operational databases, reporting platforms, and curated consumption layers while maintaining lineage, consistency, and data integrity.
- Developed reusable **PySpark and Spark SQL transformations, Delta tables, dimensional models, and Medallion Architecture layers** to optimize data preparation, curation, analytics consumption, and downstream BI workloads.
- Owned the complete development lifecycle across **solution design, cloud infrastructure, networking, security, ingestion, data preparation, modeling, CI/CD, performance tuning, deployment, monitoring, alerting, and production support**.
- Implemented DevOps and cloud-engineering practices using **Azure DevOps, Git, Terraform, Docker, CI/CD pipelines, Azure Monitor, logging, and automated deployments**, improving release consistency and operational reliability of data products.
- Provided technical leadership while collaborating with **source application teams, product owners, architects, analysts, and business stakeholders** to define data requirements, optimize Spark workloads, resolve production issues, and deliver scalable analytics solutions aligned with business objectives.

Environments: Python, SQL, PySpark, Apache Spark, Azure Databricks, Azure Data Factory, Azure Data Lake Storage, Azure Functions, Event Hubs, Azure Stream Analytics, Delta Lake, Palantir Foundry, ETL/ELT, CDC, Data Modeling, Data Migration, Azure DevOps, Terraform, Docker, CI/CD, Monitoring, Alerting, Production Support.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed cloud-based **healthcare data engineering solutions** using Python, SQL, PySpark, Spark, Azure/AWS services, and distributed processing frameworks to support analytics, operational reporting, and data-driven healthcare workflows.
- Built scalable **ETL/ELT and data-ingestion pipelines** that processed structured healthcare data from relational databases, APIs, files, and enterprise systems into cloud storage and curated analytical datasets.
- Developed Spark-based transformation workflows using **PySpark, Spark SQL, partitioning, caching, dataframes, and optimized joins**, improving processing efficiency for high-volume healthcare datasets and downstream analytics consumption.
- Designed **data models, transformation rules, reconciliation logic, and curated consumption layers** that standardized healthcare information across source systems and supported reliable BI and analytical applications.
- Implemented cloud-native integration and processing components using **AWS Lambda, Azure concepts, event-driven services, REST APIs, IAM/RBAC, storage, and secure networking patterns** across distributed healthcare applications.
- Integrated data products with **CI/CD pipelines, Git, Docker, Jenkins, GitLab, monitoring, logging, and alerting**, supporting repeatable deployments, operational observability, issue resolution, and production support.
- Collaborated with **source-system teams, product owners, architects, analysts, and business stakeholders** to define data requirements, troubleshoot pipeline bottlenecks, tune query and Spark performance, and deliver reliable analytics solutions.

Environments: Python, SQL, PySpark, Apache Spark, ETL/ELT, Data Pipelines, Data Modeling, AWS, Azure Concepts, AWS Lambda, Cloud Storage, REST APIs, PostgreSQL, MySQL, Docker, Jenkins, GitLab CI/CD, Git, Monitoring, Logging, Performance Tuning, Production Support.

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed enterprise **data platforms and analytics pipelines** using Python, SQL, PySpark, Apache Spark, and cloud-native technologies, supporting high-volume operational datasets and downstream business intelligence applications.
- Built robust **data ingestion and ETL/ELT pipelines** to extract information from relational databases, APIs, flat files, and enterprise systems, transforming raw data into standardized and curated analytical datasets.
- Developed distributed processing workflows with **PySpark, Spark SQL, dataframes, partitioning, caching, and optimized joins**, improving processing throughput and scalability across large transactional datasets.
- Designed enterprise **data models, dimensional schemas, transformation logic, aggregation layers, and warehouse structures** using PostgreSQL, MySQL, Oracle, and SQL-based platforms for reporting and analytical consumption.
- Implemented reusable data-integration services and workflows using **Python, REST APIs, event-driven architecture, batch processing, and asynchronous patterns**, connecting source applications with downstream data and analytics platforms.
- Applied DevOps practices using **Docker, Jenkins, Git, CI/CD, environment configuration, logging, monitoring, and deployment automation**, improving reliability and repeatability across development and production environments.
- Collaborated with product managers, architects, application teams, and analysts to **design data solutions, resolve ingestion and transformation issues, perform performance tuning, support deployments, and maintain production data products**.

Environments: Python, SQL, PySpark, Apache Spark, Spark SQL, Pandas, ETL/ELT, Data Ingestion, Data Modeling, PostgreSQL, MySQL, Oracle, Data Warehousing, REST APIs, Batch Processing, Docker, Jenkins, Git, CI/CD, Performance Tuning, Monitoring, Production Support.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data engineering solutions** using Python, SQL, relational databases, and distributed application technologies to support data-intensive business processes and analytics workloads across client environments.
- Built reusable **data ingestion, transformation, and batch-processing workflows** to integrate information from transactional databases, files, APIs, and enterprise applications into centralized downstream systems.
- Designed and optimized SQL-based processing using **complex joins, aggregations, stored procedures, dimensional modeling, and data-quality rules**, improving consistency and accessibility of operational datasets.
- Implemented Python components for **data cleansing, transformation, reconciliation, scheduling, workflow orchestration, and API-based data integration**, reducing manual processing and improving reliability of recurring data workflows.
- Supported cloud and distributed data architectures using **REST APIs, microservices, event-driven patterns, Linux, Docker, Git, and CI/CD**, enabling reliable movement of data across source and consumption layers.
- Implemented operational practices around **logging, monitoring, alerting, failure recovery, environment configuration, deployment support, and production incident resolution** for business-critical data services.
- Worked with architects, senior engineers, analysts, and business stakeholders to **understand source-data requirements, design scalable solutions, troubleshoot data issues, optimize performance, and support data products throughout their lifecycle**.

Environments: Python, SQL, PostgreSQL, MySQL, Data Ingestion, ETL/ELT, Data Modeling, Batch Processing, Data Transformation, Data Reconciliation, REST APIs, Microservices, Event-Driven Architecture, Docker, Git, CI/CD, Linux, Logging, Monitoring, Production Support.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.